

Part A – Introduction
Chapter A2 Ecologically Sustainable Development

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1. INTRODUCTION

Under the *Local Government Act 1993*, the consideration of Ecologically Sustainable Development (ESD) principles is a guiding principle for councils when exercising their decision-making functions.

The *Environmental Planning and Assessment Act 1979* (EP&A Act) similarly requires ESD considerations in decision-making about environmental planning and assessment.

ESD requires the effective integration of social, economic and environmental considerations in decision-making processes. For the purposes of meeting our legislative obligations, ESD can be achieved through the implementation of the following principles and programs:

a) the precautionary principle – namely, that if there are threats of serious or irreversible environmental damage, lack of full scientific certainty should not be used as a reason for postponing measures to prevent environmental degradation.

   In the application of the precautionary principle, public and private decisions should be guided by –

   i) careful evaluation to avoid, wherever practicable, serious or irreversible damage to the environment, and

   ii) an assessment of the risk-weighted consequences of various options,

b) inter-generational equity – namely, that the present generation should ensure that the health, diversity and productivity of the environment are maintained or enhanced for the benefit of future generations,

c) conservation of biological diversity and ecological integrity – namely, that conservation of biological diversity and ecological integrity should be a fundamental consideration,

d) improved valuation, pricing and incentive mechanisms – namely, that environmental factors should be included in the valuation of assets and services, such as –

   i) polluter pays – that is, those who generate pollution and waste should bear the cost of containment, avoidance or abatement,

   ii) the users of goods and services should pay prices based on the full life cycle of costs of providing goods and services, including the use of natural resources and assets and the ultimate disposal of any waste,

   iii) environmental goals, having been established, should be pursued in the most cost-effective way, by establishing incentive structures, including market mechanisms, that enable those best placed to maximise benefits or minimise costs to develop their own solutions and responses to environmental problems.

The EP&A Act seeks to promote good design and amenity of the built environment with an aim to protect the health and safety of occupants.

The DCP provisions relating to ESD are consistent with the vision and goals in our Community Strategic Plan relating to a healthy community in a liveable city.

Wollongong City Council has developed several supporting documents related to achieving ESD including:

- Sustainable Wollongong 2030: A Climate Healthy City Strategy
- Wollongong Climate Change Mitigation Plan 2023-2030
- Wollongong Climate Change Adaptation Plan 2022
- Wollongong Urban Heat Strategy 2023

- Wollongong Urban Greening Strategy 2017-2037
- Wollongong Waste and Resource Recovery Strategy 2024-2034
- West Dapto Open Space Design Manual
- West Dapto Open Space Technical Manual.

Council has an ongoing program of developing and updating supporting documents to ensure they remain relevant. Council's website should be consulted for the latest list and versions of supporting documents that may be relevant to your proposal.

2. BACKGROUND

In 2016, Council adopted the first version of Wollongong DCP Chapter A2 Ecologically Sustainable Development. This chapter forms part of the introduction to the whole DCP – it underpins controls in other DCP chapters by outlining ESD objectives that proposed developments need to consider. It also provides examples on how these objectives could be satisfied. Key focus areas to meet the principles of ESD include:

1. Climate Change Adaptation and Mitigation – seeking to ensure development is resilient to extreme weather events, rising temperatures and rising sea levels while reducing greenhouse gas emissions.

2. Waste Reduction – prioritising the use of sustainable materials, minimising construction waste, and encouraging recycling and reuse of materials.

3. Biodiversity Protection – preserving and enhancing ecosystems, green corridors and natural habitats within urban and rural development.

4. Water Sensitive Urban Design – incorporating sustainable water management techniques such as rainwater harvesting, permeable surfaces, biofiltration systems and stormwater reuse to reduce run off and improve water quality.

5. Sustainable Transport – encouraging the use of active transport options such as walking and cycling, and public transport.

6. Energy Efficiency and Renewables – optimising building design, orientation and systems to reduce energy consumption and emissions while integrating renewable energy sources such as solar photovoltaic systems and battery storage.

7. Community Health, Quality of Life, Amenity and Social Well-Being – designing accessible and inclusive and healthy environments.

8. First Nations Knowledge and Care of Country – integrating Aboriginal perspectives on sustainability, land care and ecological balance.

3. OBJECTIVES AND EXAMPLES OF HOW TO SATISFY

3.1 Climate Change Adaptation and Mitigation

Objectives

a) Ensure development incorporates climate friendly design, energy efficiency and emission reductions to increase resilience and reduce environmental impact.

b) Encourage and enable a transition to a low waste and emissions lifestyle.

c) Promote urban cooling and minimise the heat island effect.

d) Support the health of occupants by designing buildings to perform well in hot and extreme weather events.

Examples of how to satisfy

i) Prioritise passive solar design, natural ventilation and shading to reduce energy consumption.

ii) Achieve beyond baseline NABERS and BASIX ratings for energy and water in all new non-residential and residential development.

iii) Locate landscaping and tree canopy to maximise cooling of private open space and adjoining public realm.

iv) Optimise urban cooling through green infrastructure placement and street orientation to catch prevailing breezes.

v) Design subdivision design layouts in urban release areas to include appropriate allocated space for an interlocking tree canopy to be established over time.

vi) Use cool roofs, green roofs and reflective building materials to reduce the urban heat island effect.

vii) Incorporate additional site boundary setbacks to support airflow and improved building ventilation in urban areas.

viii) Making buildings all-electric.

3.2 Waste Reduction

Objectives

a) Promote the use of sustainable building materials and reduce construction and operational waste.

b) Encourage recycling and collection of organics to reduce emissions.

Examples of how to satisfy

i) Develop a Waste Management Plan for demolition and construction works to minimise landfill waste.

ii) Use construction materials that are durable, reducing the need for replacement.

iii) Provide on-site facilities for recycling and Food Organics and Garden Organics (FOGO) collection in the design of multi-unit residential and commercial development.

3.3 Biodiversity Protection

Objectives

a) Protect the unique biodiversity of the Wollongong Local Government Area.

b) Enhance urban forest by appropriate selection and siting of native flora.

Examples of how to satisfy

i) Retain and provide suitable buffers to vegetation that has existing biodiversity value and increase tree canopy coverage and native vegetation.

ii) Implement offset planting if a tree is removed by the development.

iii) Protect wildlife corridors and reconnect isolated pockets of biodiverse landscape to allow for wildlife movement in environmentally sensitive areas.

iv) Undertake works to reduce weeds and other threats to the health and viability of vegetation.

v) Provide or conserve appropriate vegetation adjacent to existing areas of biodiversity such as

riparian corridors and natural areas.

3.4 Water Sensitive Urban Design

Objectives

a) Deliver sustainable and environmentally sensitive water management practices, including rainwater harvesting, permeable surfaces and efficient stormwater treatment, to reduce potable water consumption and improve water quality and urban cooling.

Examples of how to satisfy

i) Reduce impervious surfaces and choose permeable materials to allow water to soak into the ground and reduce runoff.

ii) Install rainwater tanks and harvesting systems to collect, treat, and reuse stormwater on-site or close to the source as possible.

iii) Choose green infrastructure solutions that have lower environmental impacts and maintenance needs (for example, grassed swales and tree pit systems).

iv) Incorporate co-benefits such as waterway health, cultural values and urban amenity beyond just stormwater management.

3.5 Sustainable Transport

Objectives

a) Support the reduction of car trips and car dependance and encourage the use of sustainable transport such as active travel and public transport.

b) Encourage sustainable mixed-use transport orientated development.

c) Prioritise active transport (walking and cycling) and access to public transport to facilitate a shift away from private vehicle use.

d) Support the transition to electric vehicles, e-mobility and other low carbon transport options.

Examples of how to satisfy

i) Provide convenient and accessible on-site end-of-trip facilities (bike storage, showers, lockers) for commercial development to encourage active transport as a preferred mode of transport.

ii) Provide supporting infrastructure that enables active and low emission transport options including electric vehicle and e-bike charging facilities.

iii) Design residential subdivisions to promote active transport as the primary choice for short journeys and facilitate key connections to public transport.

iv) Design carparks to allow for suitable adaptation to alternative uses for occupants that do not rely upon vehicle ownership for their transportation needs.

3.6 Energy Efficiency and Renewables

Objectives

a) Support development that demonstrates a high level of energy efficiency through energy sourcing, storage and equitable access by occupants.

b) Maximise energy efficiency in development and use renewable energy sources to transition away

from fossil fuels.

c) Minimise the demand placed upon the existing electricity network through the use of locally generated renewable energy and storage in new development.

Examples of how to satisfy

i) Design building orientation, envelope and openings to increase energy efficiency.

ii) Include energy efficient fixtures, fittings and equipment in building specifications.

iii) Support the generation of renewable energy on-site through solar photovoltaic systems.

iv) Incorporate appropriate roof structures to accommodate renewable energy equipment and provide access for cleaning and maintenance on relevant buildings.

v) Consider spatial and equitable access requirements for renewable energy storage or other energy management systems.

vi) Enable lots within a residential subdivision to be 'all-electric'.

vii) Generate and store renewable energy on-site and minimise peak electricity demand supplied from the electricity network/grid.

3.7 Community Health, Quality of Life, Amenity and Social Well-Being

Objectives

a) Encourage and promote developments that are socially optimal and support resilience, health, and the wellbeing of communities through:

   i) improvement to indoor air quality.

   ii) equitable access to low emission energy systems and appliances.

   iii) measures to reduce the negative impacts of heat.

   iv) accessible and inclusive for people with all levels of ability.

Examples of how to satisfy

i) Rationalise energy sources to promote all-electric buildings and avoid gas supply and disconnection charges.

ii) Provide all-electric appliances for indoor cooking and heating.

iii) Ensure habitable rooms are provided with natural ventilation and cross ventilation where possible.

iv) Include ceiling fans to facilitate greater ventilation and air flow.

v) Incorporate universal building design principles for access and inclusion.

vi) Incorporate universal subdivision design principles to achieve connectivity, walkability and cycling.

vii) Development incorporates generous landscaping including a deep soil zone which supports canopy tree(s) to provide shade, natural outlook and high amenity surroundings for residents.

3.8 First Nations Knowledge and Care of Country

Objective

a) Consider Aboriginal knowledge, cultural values and sustainable land management practices during

the project design phase.

Examples of how to satisfy

i) Undertake appropriate due diligence for projects located near or on an Aboriginal significant site.

ii) Demonstrated engagement with Aboriginal people in a culturally appropriate way.

4. HOW TO USE THIS CHAPTER

When determining a development application, a consent authority is required under section 4.15 of the EP&A Act to consider the likely impacts of that development, including environmental impacts on both the natural and built environments, the social and economic impacts in the locality and the public interest.

Council expects all new development in the Wollongong Local Government Area is consistent with the principles of ESD and resilient to climate change.

This chapter outlines key focus areas and objectives, and provides examples of how to satisfy the ESD objectives which proponents for new developments must consider and address as part of their development application. The examples listed in this chapter are not exhaustive and proponents are encouraged to explore other design features relevant to their particular site and development which would result in more sustainable outcomes for both the occupants of the development and the wider community.

This chapter provides context for specific controls contained in other DCP chapters. By reviewing their application against the how to satisfy examples presented in this chapter, applicants can identify elements of their proposal that are consistent with the principles of ESD.

Development controls for ESD have been integrated into relevant sections of this DCP. This chapter must be read in conjunction with other relevant chapters.

5. RELATIONSHIP WITH OTHER PLANS AND POLICIES

This chapter reflects recent developments in federal and state government policies, as well as the strategic direction of Wollongong City Council. This chapter must be read in conjunction with other relevant DCP chapters. ESD principles and objectives are considered in each update of the Wollongong DCP.